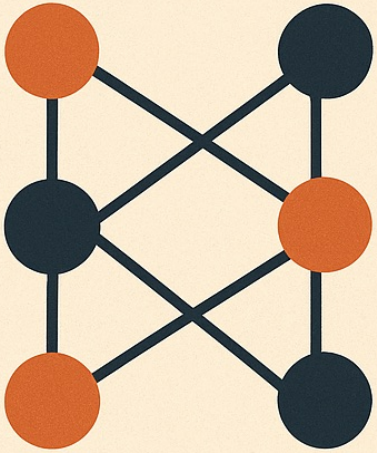
INTRO TO MACHINE LEARNING

AND NEURAL NETWORKS





Lecture 4: Linear Regression - Part III

#Agenda

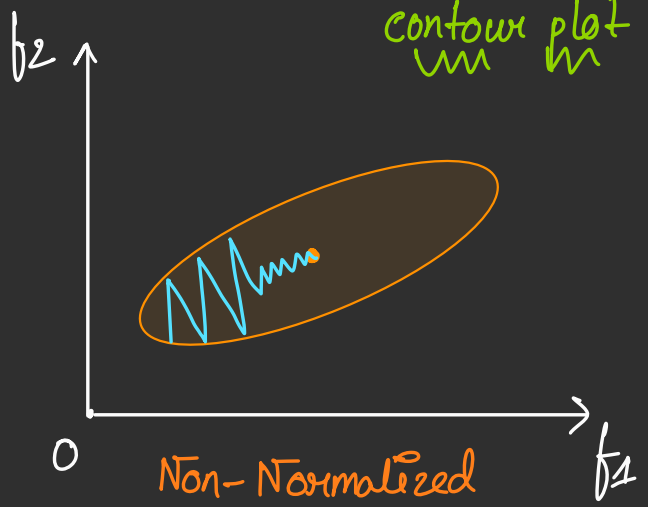
1. Feature Scaling
2. Adjusted R-Squared
3. Stats Model
4. Assumption of Linear Regression

#Feature Scaling

↳ process of doing Normalization / Standardization of features

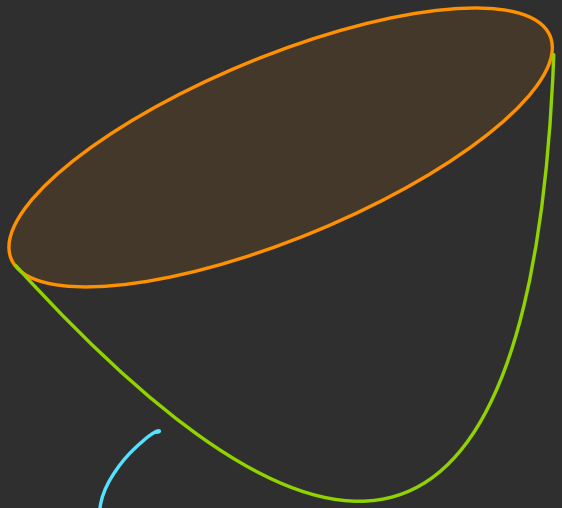
Yes, we have done it.

(To avoid any bias in my result)



contour plot

Non-Normalized feature



Loss Fun^n = MSE

$(y - \hat{y})^2$

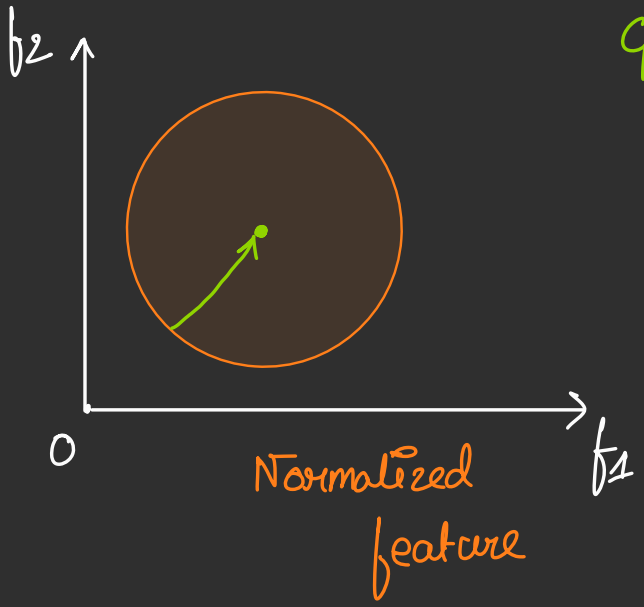
GD takes time

↳ Training of model will take long

$$var = \underbrace{(x_i - \bar{x})^2}$$

↳ unit value

$$std = \sqrt{var}$$



GD will takes
less time → training
faster

Adjusted R_2 Score

↳ What happened to R_2 score? What is the problem with R_2 score?

$$R_2 = 1 - \frac{SS_m}{SS_T} = 1 - \frac{\text{Your model MSE}}{\text{Mean model MSE}}$$

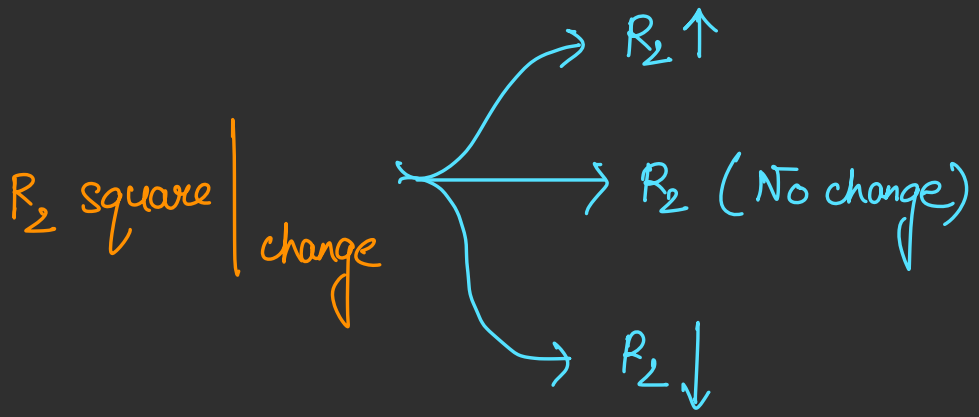
$\hat{y} = w_1 x_1 + w_2 x_2 + w_0$

↳ $R_2 = z_1$

$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0$

↳ $R_2 = z_2$

When I added a new feature.



1) The feature → Relavant

R_1

(Can't Say)

2) The feature → Irrelavant

R_2

(Can't Say)

Adjusted R_2 Score

$$\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_0$$

$$\text{Adj } R_2 \text{ score} \Rightarrow 1 - \left\{ \frac{(1-R_2)(m-1)}{(m-d-1)} \right\}$$

$1000 - 100 - 1$

$900 - 1$

899

- $m \longrightarrow$ No of rows / No of dt pts
- $d \longrightarrow$ No of feature

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + w_5 x_5 + w_0$$

$w_4 x_4$ — Imp

$w_5 x_5$ — No Imp

$R^2 \uparrow$ Adj R^2 Score $\downarrow$

R_2 score $\longrightarrow (-\infty, 1]$

Adj° R_2 score $\longrightarrow (-\infty, 1]$

$R^2 \Rightarrow 1 - \frac{SS_{res}}{SS_{total}}$

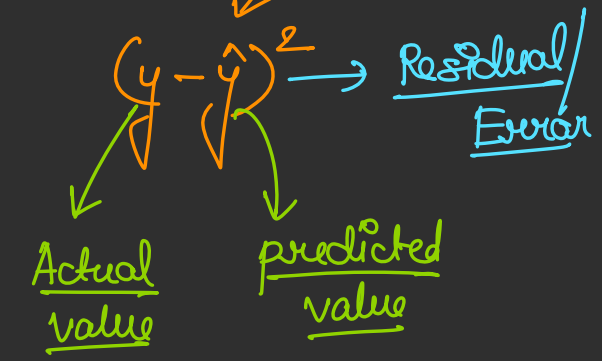
SS_{res} ↑ very large number

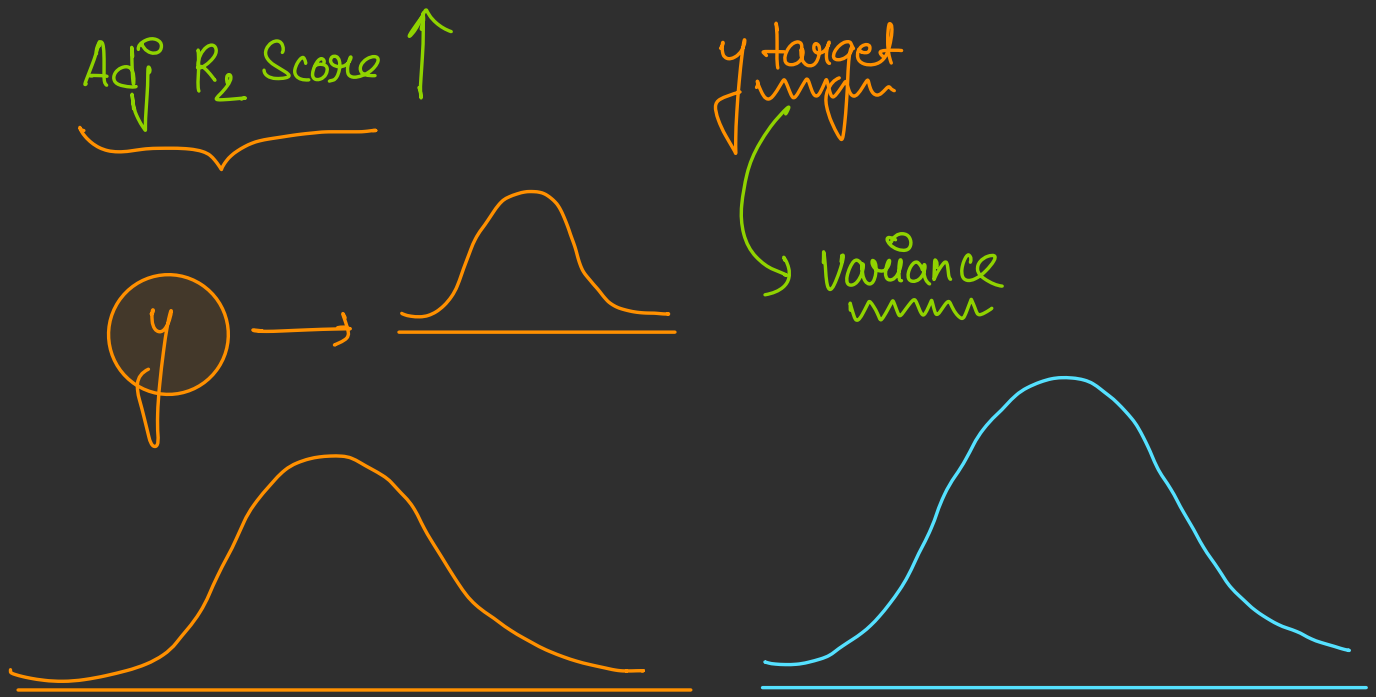
SS_{total} ↓ very small number

$\Rightarrow -\infty$ { Worst than Baseline }

$R^2 \Rightarrow -\infty$

R^2 score = $1 - \frac{\text{Your model MSE}}{\text{Mean model MSE}}$




Adj R_2 Score ↑
y target
Variance
y

Stats Model

OLS → "Ordinary Least Square"

↳ Mathematical way to represent linear reg

↓

estimate the parameters (w, w_0) by minimizing the sum of square residuals

MSE

OLS

↳ Linea Algebra to solve and find parameters

Sklearn

Linear Regression

↓

gradient descent

↓

to minimize my loss & to get optimize value of w, w_0

Stats Model

Linear Rege

↓

Linear algebra (Mat Mult, dot prod)

↓

to get the optimize value of w & w_0

OLS

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots w_n x_n + w_0 = w^T x + w_0$$

$$\hat{y} = Xw + w_0$$

$$\text{Loss} \Rightarrow \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \| y - (Xw + b) \|^2$$

Closed form Solution

$$w = (X^T \cdot X)^{-1} \cdot X^T \cdot y \quad (\text{OLS solution})$$

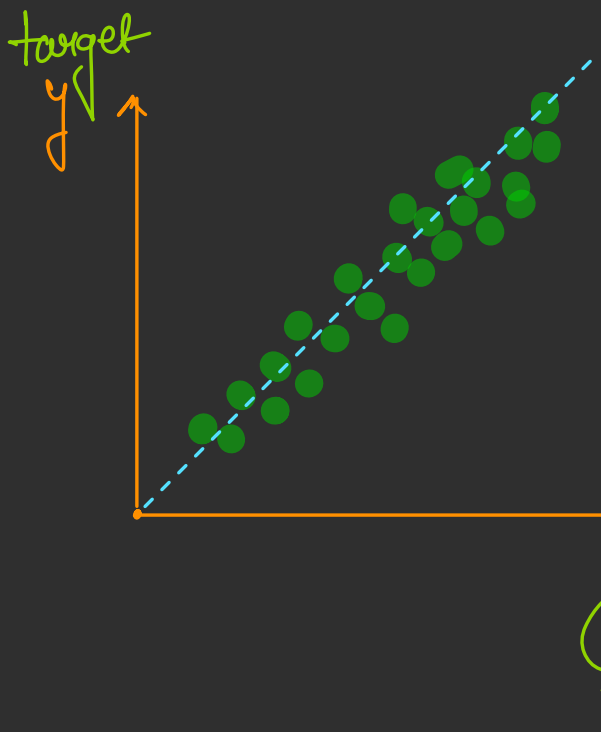
OLS → Faster for smaller data

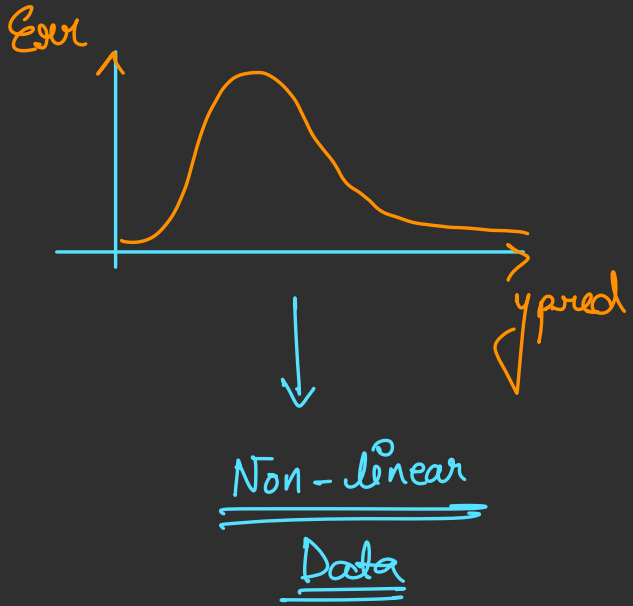
Sklearn → Used for larger data

Assumptions of Linear Regression

1) Assumption of linearity
2) No multicollinearity
3) Normality of Residuals
4) No Hetroskedasticity
5) No Auto correlation

Assumption of linearity



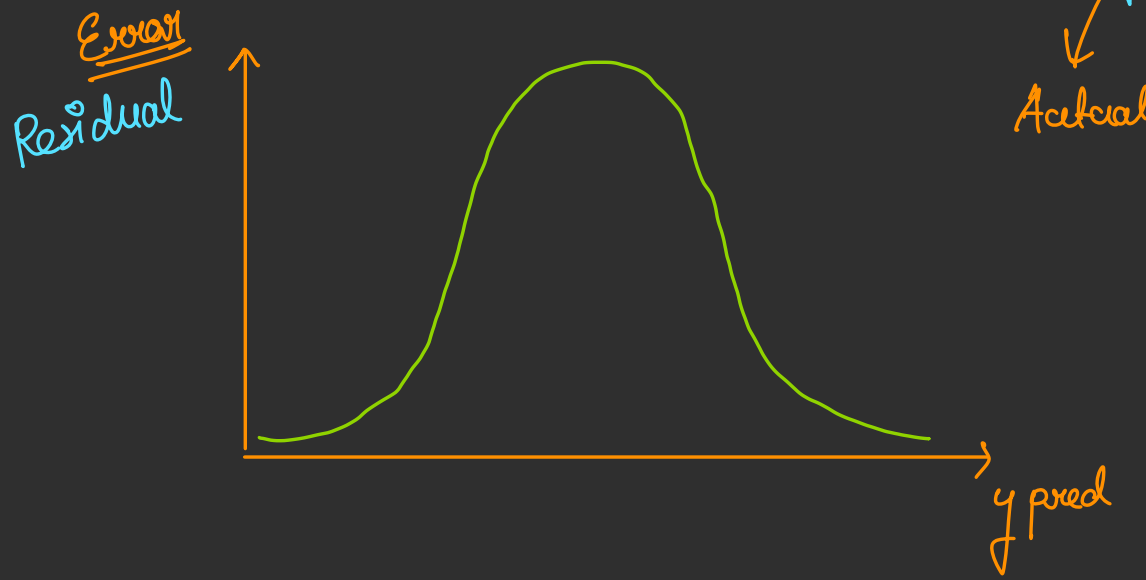


Normality of residual

error = $(y - \hat{y})$

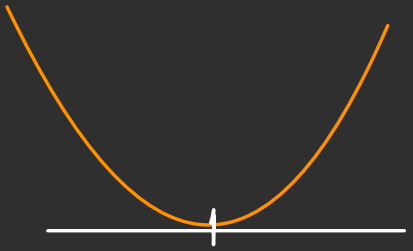
Actual

predicte



Building your Loss Function

- MAE ✓
- MSE ✓
- RMSE ✓



Loss fun^n

W, W0

Reg

Fixed set of loss

